

⌈ Preliminary.

Lemma. Let  $T: V \rightarrow V$  be a linear map on finite-dimensional space  $V$ , and let  $W \subseteq V$  be a  $T$ -invariant subspace.

Suppose  $v_1, \dots, v_k$  are eigenvectors of  $T$  corresponding to distinct eigenvalues.

If  $v_1 + v_2 + \dots + v_k \in W$ , then  $v_i \in W$  for all  $1 \leq i \leq k$ .

Proof. If  $k=1$ , trivial. Suppose that for any integer smaller than  $k$ , the statement holds.

Assume  $v_1, \dots, v_k$  are eigenvectors of  $T$  corresponding to distinct eigenvalues  $\lambda_i$ .

If  $v_1 + \dots + v_k \in W$ , then

$$T(v_1 + \dots + v_k) = \lambda_1 v_1 + \dots + \lambda_k v_k \in W \quad \text{and} \quad \lambda_1(v_1 + \dots + v_k) \in W$$

$\uparrow$   $\uparrow$   
 $W$  is  $T$ -invariant      Closed under scalar multiplication.

Subtracting the vectors:

$$\sum_{i=1}^k (\lambda_i - \lambda_1) \cdot v_i = \sum_{i=2}^k (\lambda_i - \lambda_1) \cdot v_i \in W$$

$i \neq k \Rightarrow \text{not zero.}$

Therefore,  $v_i \in W$  for all  $1 \leq i \leq k-1$ , by the hypothesis. Moreover,

$$v_k = v_1 + \dots + v_k - (v_1 + \dots + v_{k-1}) \in W.$$

Thm. If  $T: V \rightarrow V$  is diagonalizable, then for any  $T$ -invariant  $W \subseteq V$ ,

$$T|_{W}: W \rightarrow W \text{ is } T|_W$$

is also diagonalizable.

Proof. Let  $\beta$  be a basis for  $V$  s.t.  $[T]_\beta$  is diagonal.

Let  $\beta'$  be a basis for  $W$ . Then, every vector of  $\beta'$  is represented by linear combination of  $\beta$ . That is, if  $\beta = \{v_1, \dots, v_n\}$  and  $\beta' = \{w_1, \dots, w_k\}$ , then for all  $1 \leq j \leq k$

$$w_j = a_{1,j} v_1 + \dots + a_{n,j} v_n \quad (a_{i,j} \in F).$$

Consider a  $n$  by  $k$  matrix  $F^k \rightarrow F^n$

$$\begin{pmatrix} [w_1]_\beta & \dots & [w_k]_\beta \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & - & - & - \\ \vdots & - & - & - \\ a_{n1} & - & - & - \end{pmatrix}$$

has rank as  $k$ .

### Simultaneously Diagonalizable.

In this section,  $V$  is finite-dimensional vector space over  $F$ , and  $T_1, T_2, \dots: V \rightarrow V$  are linear operators.

Def. Linear operators  $T_1$  and  $T_2$  are called simultaneously diagonalizable if:

there exists a basis  $\beta$  for  $V$  s.t. both  $[T_1]_\beta$  and  $[T_2]_\beta$  are diagonal.

Lemma. If  $T_1, T_2$  are simultaneously diagonalizable by a basis  $\beta$ , then for any basis  $\alpha$ ,  $[T_1]_\alpha$  and  $[T_2]_\alpha$  are simul. diagonalizable.

Proof. Clearly,  $[T_i]_\beta = [T_i]_\alpha^\beta [T_i]_\alpha [T_i]_\beta^\alpha$  thus obtain the result.

Lemma. If  $T$  is diagonalizable, then  $T^m$  and  $T$  are simultaneously diagonalizable, for any  $m \geq 1$ .

Proof. Let  $\beta$  be a basis for  $V$  satisfying  $[T]_\beta$  is diagonal. Then

$$[T^m]_\beta = [T]_\beta^m \text{ diagonal.}$$

Thm. If  $T_1, T_2$  are simultaneously diagonalizable by a basis  $\beta$ , then  $T_1$  and  $T_2$  commute. That is,  $T_1 \circ T_2 = T_2 \circ T_1$ .

Proof.  $[T_1 \circ T_2]_\beta = [T_1]_\beta \cdot [T_2]_\beta = [T_2]_\beta \cdot [T_1]_\beta = [T_2 \circ T_1]_\beta$ .

↕ Converse

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_n \end{pmatrix} \cdot \begin{pmatrix} \mu_1 & 0 \\ 0 & \mu_n \end{pmatrix} = \begin{pmatrix} \lambda_1 \mu_1 & 0 \\ 0 & \lambda_n \mu_n \end{pmatrix}$$

Thm. If  $T_1$  and  $T_2$  are diagonalizable and commute, then  $T_1$  and  $T_2$  are simultaneously diagonalizable.

Thm. Let  $\mathcal{C}$  be a collection of diagonalizable linear operators on  $V$ .

there exists a basis  $\beta$  for  $V$  s.t.  $[T]_\beta$  is diagonal for all  $T \in \mathcal{C}$

if and only if

$\mathcal{C}$  commute under composition.



